

Data Unloading



Learn2
CloudData

Powered by

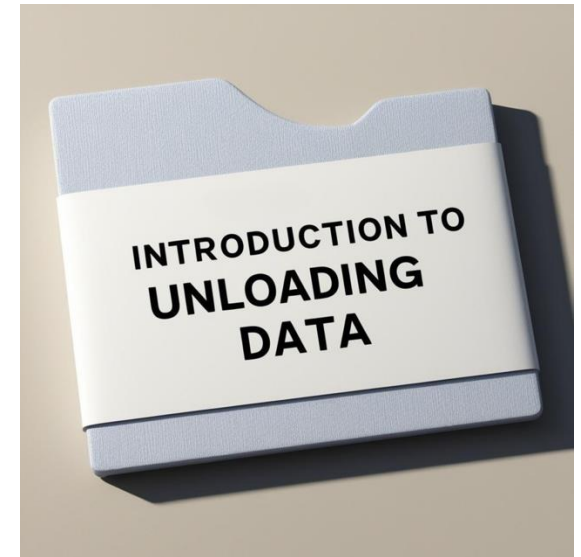
www.learn2clouddata.com

Sujith Nair

Cloud Data Architect
Snowflake Snowpro Certified

Section Introduction–Unloading Data

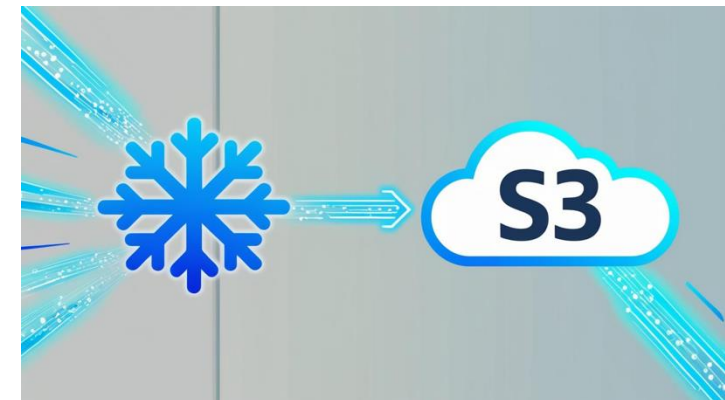
- What is Data Unloading ?
- Data Unloading Use Cases ?
- Data Unloading Features ?
- LAB-1-Unloading files to AWS s3 from Snowflake
- LAB-2-Unloading files with COPY Option SINGLE=TRUE ,MAX_FILE_SIZE, OVERWRITE=TRUE
- LAB-3-Unloading files with data from more than 1 table
- LAB-4-Partitioned Data Unloading
- LAB-5-Unloading JSON data
- LAB-6-Unloading parquet data
- Summary



What is Data Unloading?

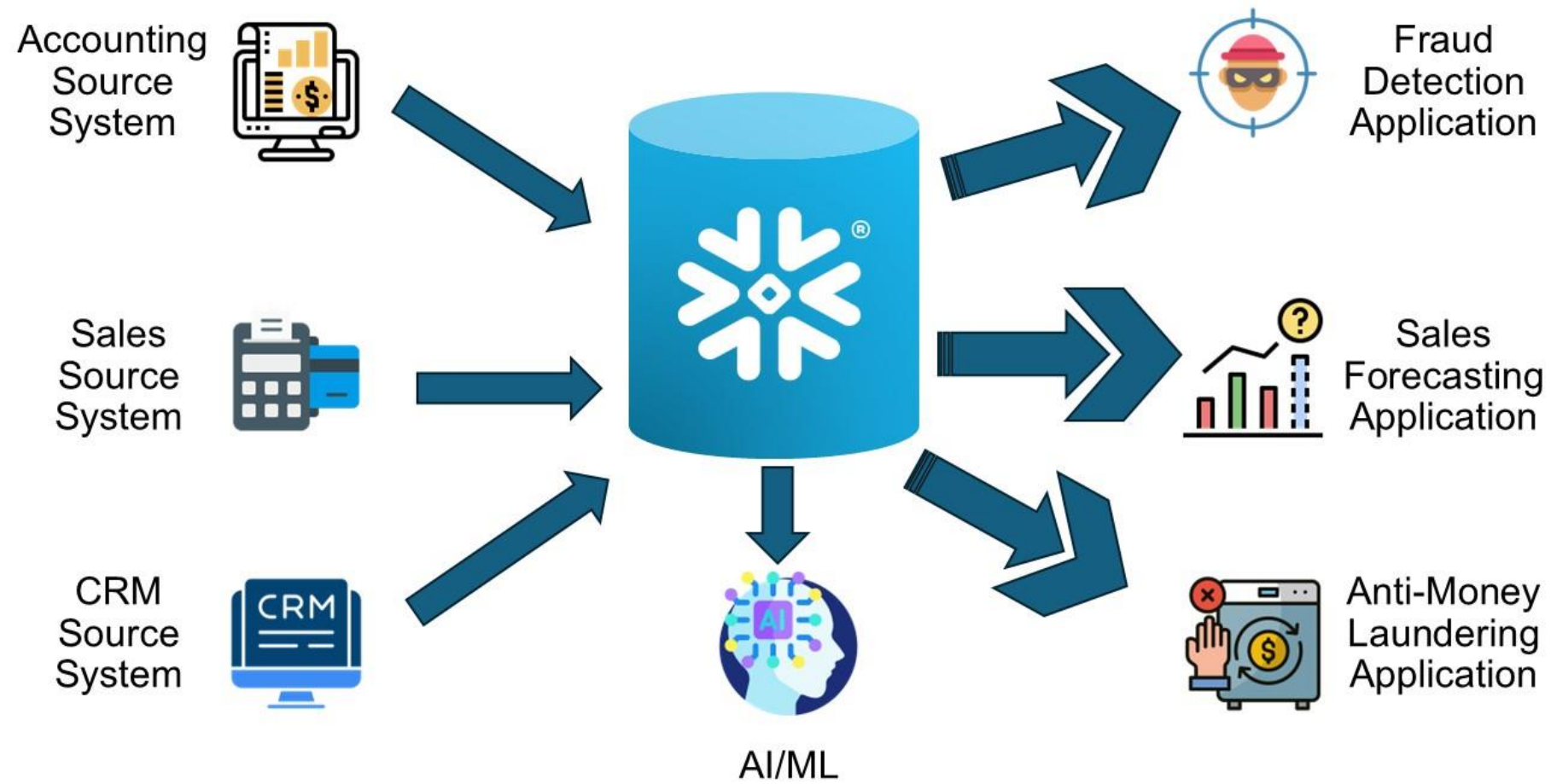
Data Unloading in Snowflake refers to the process of exporting data also known as unloading data from Snowflake tables to external storage such as

- Amazon s3
- Google Cloud Storage
- Microsoft Azure Blob Storage



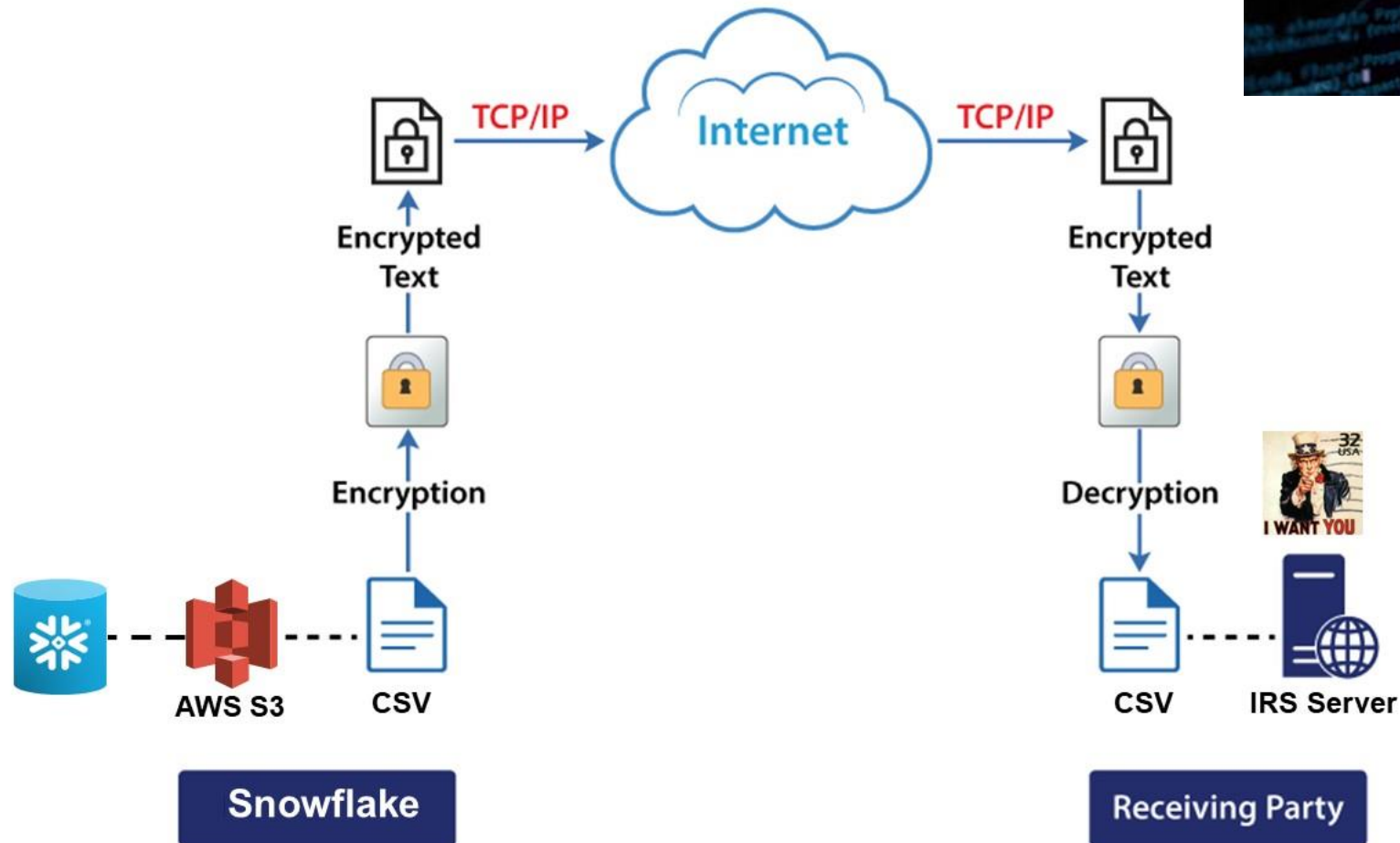
Data Unloading Use Case

Use by External Applications



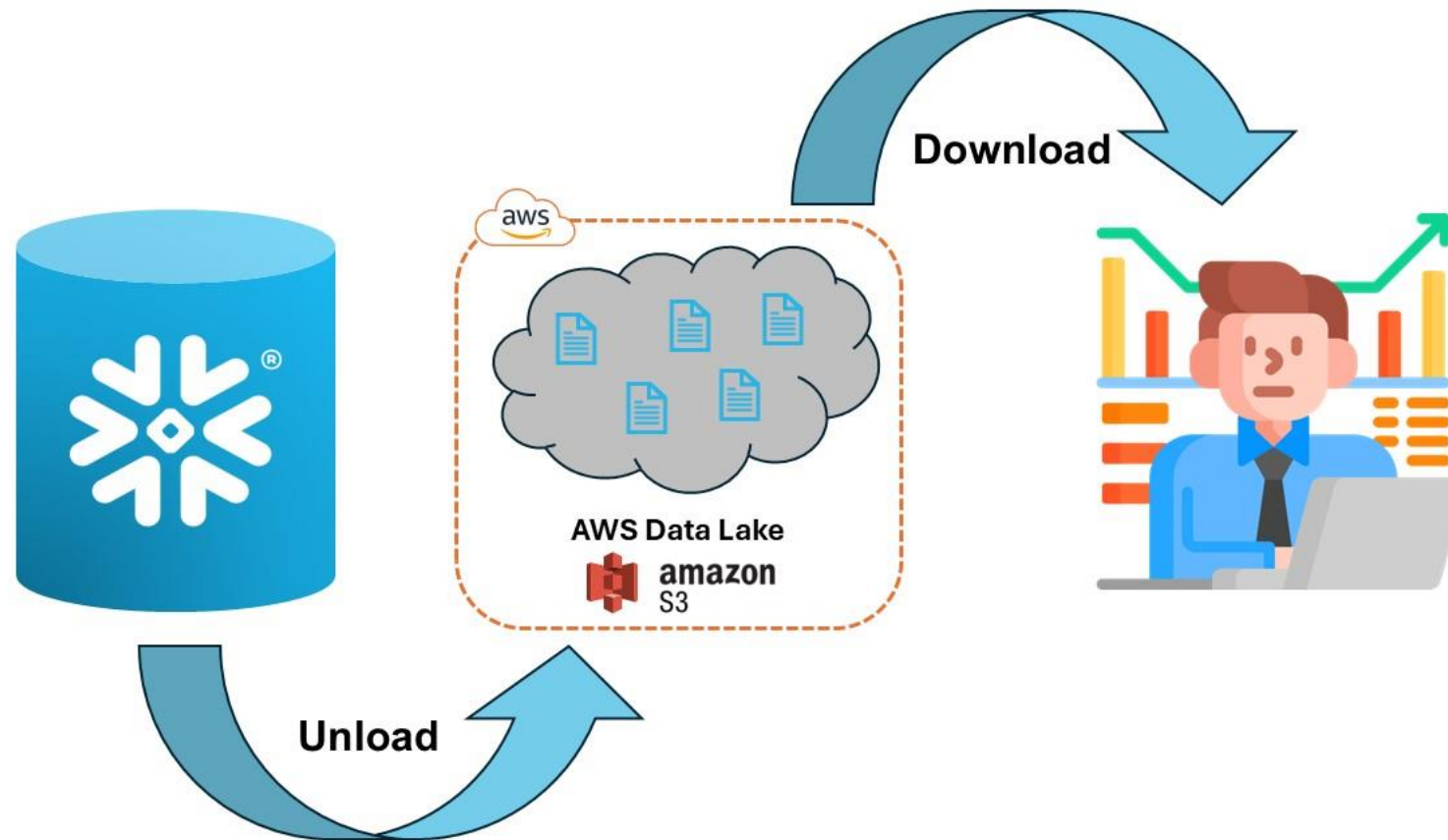
Data Unloading Use Case

Data Feed for 3rd Parties



Data Unloading Use Case

Share Table Data with Business Users



What is Data Unloading

Features of Data Unloading

- Unloading Data uses the COPY statement.
- Join multiple tables in the COPY statement
- Unloading Data supports COPY options
 - OVERWRITE =TRUE
 - SINGLE =TRUE|FALSE
 - HEADER =TRUE|FALSE
 - MAX_FILE_SIZE
 - COMPRESSION = **GZIP | BZ2 | NONE | AUTO**
- Data can be unloaded in Snowflake in the following formats.
 - CSV
 - Parquet
 - JSON
 - ORC
 - XML
 - Avro



Section Summary

- Data Unloading is the process of exporting data also know as unloading data from Snowflake tables into external cloud storage such as AWS, Azure or GCP
- Data is Unloaded from Snowflake for use in External Applications, Data Feed for 3rd parties or to share Snowflake data with business users to name a few
- Data Unloading supports several COPY options like
 - SINGLE = TRUE|FALSE
 - HEADER=TRUE|FALSE
 - OVERWRITE =TRUE|FALSE
 - MAX_FILE_SIZE
 - COMPRESSION
- SINGLE=TRUE forces the unload COPY statement to create a single file as output instead of multiple files which is its default behavior
- By default unload COPY uses parallelism to split table data into multiple files to improve unload performance
- USE HEADER=TRUE to create files with headers in them
- USE OVERWRITE=TRUE to overwrite a file that is already present on the external stage
- Size of a file can be controlled with MAX_FILE_FIZE parameter
- The default MAX_FILE_SIZE is 16 MB for a file generated with unload COPY. This can be overridden by setting MAX_FILE_SIZE to a value different from 16 MB eg: MAX_FILE_SIZE =250 MB
- Files are compressed by default when they are unloaded from Snowflake

Section Summary

- USE COMPRESSION =**GZIP | BZ2 | NONE | AUTO** to set the compression applied to the exported file
- Multiple tables can be joined in the unload COPY SELECT statement to produce a files with data from multiple tables. SQL function can be used in the SELECT statement to format the data.
- A view can be used to replace a complex SQL statement and use it in the Unload COPY.
- Files can be unloaded with the **PARTITION BY CLAUSE** <column-name> to break the files into multiple partitions on s3.
- SQL function and expressions can be used in the PARTITION BY CLAUSE eg: **YEAR(HIRE_DATE)**
- Partitioned data folders are used in Data Lake storage to improve partition pruning and query performance.
- Provide a file name prefix to the unload COPY to avoid use of generic data file names in the file output
- Data can be unloaded in Snowflake in the following formats.
 - CSV
 - Parquet
 - JSON
 - ORC
 - XML
 - Avro

SECTION SUMMARY FOR UNLOADING DATA

Thank you



Learn2
CloudData

Powered by

www.learn2clouddata.com

Find Us

www.learn2clouddata.com

+01 234 567 89